# AGRICULTURE
# INDUSTRY REVIEW

**Rowan Das**
October 2026

## Opening Statement

Agriculture is one of the oldest industries in the world, and almost every economy still runs on it. Farmland has held its value for centuries, which is why pensions and endowments still own it. Most farmers are price takers. They don't get to set what they pay for seed, fertilizer and fuel, or what they sell their crop for, so profits come down to input costs and when they decide to sell. Farming also takes a lot of capital. Land and equipment last for decades and are mostly bought with debt, so the industry gets hit hard by changes in interest rates, government policy, trade and currency. AI is starting to help, with tools like yield forecasting, satellite crop tracking and precision equipment that let farmers grow more per acre and waste less.

## Market Exposure

**Pure-play Agriculture & Inputs (~C$190bn).** Fertilizer, seed and crop protection producers such as Nutrien, Mosaic, CF Industries and Corteva. These names carry the most direct leverage to crop prices and farm spending.

**U.S. Ag Products & Services (~US$78bn).** Grain merchants and processors such as ADM and Bunge, which earn mainly on volumes and processing margins rather than outright crop prices.

**Machinery & Ag-Tech.** Deere, CNH Industrial and AGCO, plus a growing set of venture-backed ag-tech firms. This is the fastest-growing part of the sector, though equipment demand follows farm income.

## Key Macroeconomic Drivers

### Interest Rates

Farming is one of the most capital-intensive industries, so borrowing costs flow straight to margins. Higher rates raise the cost of land, equipment and operating loans and make bonds more attractive relative to farmland, which pressures land values. U.S. farm interest expense is forecast at a record $33bn+ in 2026, absorbing ~21% of operating earnings vs. 12% in 2021.

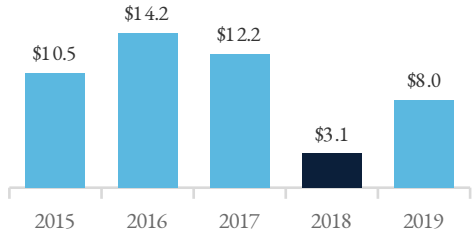
### Fiscal Policy

Policy decides how easily capital reaches farmland. Australia is relatively open to outside buyers, while provinces such as Saskatchewan and Quebec restrict non-resident ownership, shrinking the buyer pool. Government support has historically cushioned incomes in down years: the U.S. paid $23bn in trade aid for 2018-19, and direct payments are forecast at $47.4bn in 2026.

*(US$ unless noted)*

| Key Stats | |
|---|---|
| **U.S. Farm Economics (2026E)** | |
| Net Farm Income | $158.4bn |
| Direct Government Payments | $47.4bn |
| Interest Expense | $33bn+ |
| Interest / Operating Earnings | 21% |
| *Same Ratio in 2021* | *12%* |
| Total Farm Debt | $605.1bn |
| **Debt / Assets** | **13.5%** |
| **Trade** | |
| China Soy Pledge (to 2028) | 25 MMT/yr |
| Soy Exports to China, 2017 | $12.2bn |
| Soy Exports to China, 2018 | $3.1bn |
| **Change, 2017 to 2018** | **(74%)** |

U.S. Soybean Exports to China (US$bn)




*2018: China's 25% retaliatory tariff; Brazil took share.*

### Tariffs and Trade Policy

Trade barriers strand supply at home, push domestic prices down and hand share to rival exporters. The 1980 Soviet grain embargo cancelled ~17 MMT of U.S. sales and let Argentina take share. The pattern repeated in 2018, when China's 25% retaliatory tariff cut U.S. soybean exports to China by 74% and Brazil captured the business. China's pledge to buy 25 MMT a year of U.S. soybeans through 2028 helps, but history shows how quickly access can reverse.

### Foreign Exchange

Grain trades globally in U.S. dollars, so a weaker home currency lifts exporters' local-currency revenue and competitiveness. A soft Canadian dollar supports Canadian growers and fertilizer producers selling into U.S. dollar-priced markets; a stronger loonie is a headwind.

### The 1980s Farm Crisis: When the Drivers Turn Together

The early 1980s show what happens when rates, policy and trade all move against farmers at once. The prime rate hit 21.5% in 1981, U.S. farm exports fell more than 20% between 1981 and 1983 after the Soviet embargo, and farm debt doubled to $215bn by 1984 from 1978. Farmland values fell as much as 60% in parts of the Midwest between 1981 and 1985, and Massey Ferguson, Allis-Chalmers and International Harvester each lost more than $200M in 1982.

The lesson for investors is pass-through: growers absorb most of a macro shock, while the input and equipment suppliers with pricing power can push part of it onto customers.

# AGRICULTURE
# KEY INDUSTRY TRENDS

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## Industry Outlook

World agricultural and fish production is projected to grow 13% from 2026 to 2035, mostly through higher yields rather than new farmland (OECD-FAO), after the value of global production rose 84% between 2000 and 2021 to US$3.7 trillion. In 2026, large crops are meeting much higher fuel and fertilizer bills, so margins depend on using technology to raise yields and cut inputs.

## Key Industry Trends

### Precision Agriculture and Drones

GPS-based technologies were used on about 40% of U.S. farm and ranch acreage by 2019, and auto-steer guidance on 58% of corn acres by 2016. Spray drones are the fastest-growing tool, treating about 16.4 million U.S. acres in 2025, up 58.7% from 2024, as growers target applications to the parts of a field that need them.

### Restoring Soil Health

U.S. cover crop acreage reached 18.0 million acres in the 2022 Census of Agriculture, up 17% from 2017 and roughly 75% from 2012. Cover crops and reduced tillage limit erosion and build organic matter, which can lower fertilizer needs over time.

### Precision Harvesters

Combines pair yield monitors with GPS to map output across each field and set next season's seeding and fertilizer plans. Upgrades are being deferred, though: U.S. combine sales fell 9.1% and tractor sales 12.4% through August 2026.

### Yield per Acre

U.S. corn yields have doubled since 1980, from 91 to a record 186.5 bushels per acre in 2025, a trend gain of just under 2 bushels a year, and soybeans set a record of 53.0. Most of the gain comes from better seed genetics, supported by fertilizer and precision management. The U.S. harvests about twice the world average per acre and twice Brazil's, so output grows mainly through yield rather than new land. Weather still matters: USDA cut its 2026 corn forecast to 178.5 bushels in September.

### Output Expansion and Gross Production Value

USDA forecasts a record 2026-27 U.S. soybean crop of 4.54 billion bushels and 15.8 billion bushels of corn, after a record 17.0 billion in 2025-26; Brazil's soybean crop is forecast at 186 MMT. Canadian farm cash receipts hit a record C$101.4bn in 2025, and Australia's gross value of production is forecast at A$99.4bn for 2026-27, 5% below its 2025-26 record.

### Input Costs Outpacing Revenue

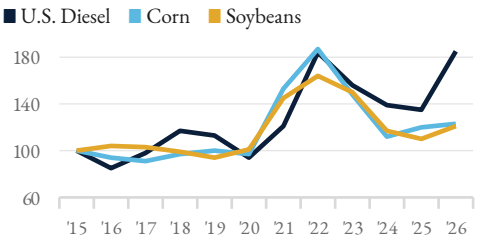
USDA forecasts 2026 U.S. production expenses up 4.5% to $492.8bn while cash receipts slip 0.3%, led by fuel and oil (+28.8%) and fertilizer (+15.3%). Diesel averaged $5.00 per gallon through September, up 37% on 2025, while corn rose only 3%, and Canada's ~$20bn of retaliatory tariffs from September 8 now cover combines and header components.

Diesel and crop prices usually move together (a 0.84 correlation with corn over 2014-2025, 0.82 with soybeans) because energy feeds into fertilizer, freight and biofuel demand. In 2026 the link broke: diesel is back near its 2022 level while corn sits 34% below it.

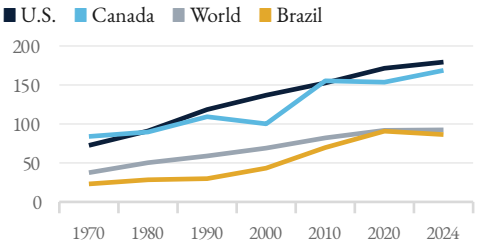
Subsector Growth (U.S., 2026F YoY)

| | |
|---|---|
| **Row Crops (cash receipts)** | |
| Corn | +11.3% |
| Soybeans | +10.0% |
| Cotton | +12.5% |
| Wheat | (2.5%) |
| **Perennial & Specialty Crops** | |
| Vegetables & Melons | +15.0% |
| Fruits & Tree Nuts | (0.4%) |
| **Livestock, Meat & Dairy** | |
| Cattle & Calves | +5.2% |
| Hogs | (4.0%) |
| Broilers | (2.8%) |
| Eggs | (66.3%) |
| Milk | (4.3%) |
| **Machinery (unit sales, YTD Aug)** | |
| Total Farm Tractors | (12.4%) |
| 4WD Tractors | (24.4%) |
| Self-Propelled Combines | (9.1%) |
| **Services** | |
| Drone Application Acres (2025) | +58.7% |

Diesel vs. Crop Prices (2015 = 100)




Corn Yield per Acre (bushels)




Regional Performance

**United States.** Net farm income forecast down 2.6% to $158.4bn in 2026 as expenses (+4.5%) outrun receipts.

**Canada.** Record C$101.4bn of 2025 cash receipts (+3.4%), but net income flat at C$8.3bn as canola exports slid.

**Brazil & Argentina.** Soybean crops of 186 and 50 MMT forecast for 2026-27; Brazil gains whenever U.S.-China trade breaks down.

**Australia.** 2026-27 gross value forecast at A$99.4bn, 5% below the record, with broadacre farm profit down 39%.

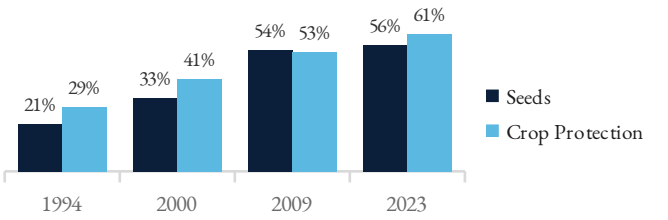
# AGRICULTURE
# KEY INDUSTRY PLAYERS

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October 2026

## Competitive Landscape

The top four firms in seeds, crop protection, machinery and animal health hold 41% to 61% of global sales, up from 21% to 32% in 1994. They outspend smaller rivals on R&D, hold the key patents and control dealer and distribution networks, which makes it hard for new competitors to win share and gives the leaders pricing power. Their stocks behave more like industrial names than farm names, moving with farm income but holding market share and margins through downturns.

Top-4 Global Market Share Over Time (%)




Top-4 Share of Global Sales by Segment, 2023 (%)




*Concentration keeps rising where R&D, patents and regulatory approval are the barriers to entry. 1994-2009: USDA ERS; 2023: ETC Group & GRAIN.*

### Machinery: Deere & Company (NYSE: DE)

A staple of North American farming since John Deere's 1837 steel plow, Deere is one of the few equipment makers to come through the 1980s farm crisis intact. While Massey Ferguson, Allis-Chalmers and International Harvester each lost over $200M in 1982 and were later merged or sold, Deere had used its 1970s earnings to cut debt and modernize, kept interest and overhead under 5% of sales (14% at its nearest rival) and supported its dealers through the downturn. Today it is buying into ag-tech: Blue River Technology ($305M, 2017), Bear Flag Robotics ($250M, 2021), and Sentera and GUSS Automation (2025). Earnings hold up through the cycle, with FY2025 net income of $5.0bn on $45.7bn of revenue, a 15.4% margin in Production & Precision Ag, and $890M from its captive finance arm.

JOHN DEERE

| | |
|---|---|
| Deere | **15%** |
| CNH Industrial | **10%** |
| AGCO | **8%** |
| Kubota | **8%** |

*Share of 2023 global sales*

### Seeds & Crop Protection: Corteva (NYSE: CTVA)

Bayer, Corteva, Syngenta and BASF control 56% of global commercial seed sales and 61% of pesticides. The moat is patented genetics and chemistry that take years of R&D and regulatory approval, plus hybrid seed that farmers must buy again every season. Corteva, spun out of DowDuPont in 2019, owns Pioneer and keeps building through M&A, adding biologicals makers Stoller and Symborg in 2023. It is now splitting in two: the seed and genetics business spins off as Vylor in Q4 2026, while Corteva keeps crop protection and digital. Seed is the bigger engine, with $7.6bn of first-half 2026 sales vs. $3.7bn in crop protection.

CORTEVA

| | |
|---|---|
| Big 4: Seeds | **56%** |
| Big 4: Pesticides | **61%** |

*Combined share of global sales*

### Animal Health: Zoetis (NYSE: ZTS)

Zoetis, Merck Animal Health, Boehringer Ingelheim and Elanco hold about half of global animal pharma sales. Products need regulatory approval, are sold through veterinarians and are protected by patents on brands such as Apoquel and Simparica Trio. Zoetis, spun out of Pfizer in 2013, leads with 2025 revenue of $9.5bn (+6% organic) and $2.7bn of net income, a ~28% margin, with about 70% of sales from less cyclical companion animals. In livestock, sustainability is a new growth path: Elanco's Experior is approved in the U.S. to reduce ammonia emissions from cattle.

zoetis

| | |
|---|---|
| Zoetis | **18%** |
| Merck (MSD) | **12%** |
| Boehringer | **11%** |
| Elanco | **9%** |

*Share of 2023 global sales*

### Ag Biotech & Biologicals: Open to New Entrants

Biotech is the one segment no group dominates yet. Gene-editing and microbial inputs are being built by venture-backed firms such as Pivot Bio and Pairwise, and the market is still small: biologicals were $156M, about 1.4%, of Corteva's first-half 2026 sales. Expect the seed playbook to repeat, with the majors buying proven startups once their products clear regulators.

**No dominant player**

*Room for entry while the incumbents' biologicals arms are still small*

*Source: ETC Group & GRAIN, top agribusiness report (Jun-2025, 2023 sales); USDA ERS Amber Waves (Dec-2012); Deere FY2025 results (Nov-2025); Farm Equipment (2018, 2023); AgFunder News (2021); Agri Navigator (Aug-2025); Corteva (Oct-2025) and Q2-2026 10-Q; DTN (May-2026); Zoetis FY2025 results.*

# AGRICULTURE
# DEAL SPOTLIGHT

**Rowan Das**
Rotman Commerce Student Fund
October 2026

## Pacific Avenue's $1.1bn Carve-out of GardenCore

In May 2026, Pacific Avenue Capital Partners bought Oldcastle Lawn & Garden from CRH for $1.1bn out of its Fund II and renamed it GardenCore. The company makes mulch, soil, stone and lime products at 55+ U.S. plants with 1,400+ employees, selling to home improvement retailers and garden centers. Jefferies and Weil advised the buyer, with Sidley on the debt financing, and BofA Securities ran the sale for CRH. The financing shows how a sponsor can buy a plant-heavy business without stretching on bank debt.

### How the Deal Was Funded

The largest single source of cash was real estate. At closing, GardenCore sold all 43 of its owned plants to W. P. Carey for $400M and leased them back on a 20-year triple-net lease with fixed annual rent increases. That covered about 36% of the price, so debt and equity only had to fund roughly $700M. The debt terms were not disclosed; we model a $175M ABL revolver ($75M drawn) for the seasonal inventory build and a $375M first lien private credit term loan at SOFR + 5.25%, leaving an estimated $298M equity check, about 18% of Fund II.

### Why the Sale-Leaseback Matters

At an assumed 7.4% cap rate, the lease costs about $29.6M a year, cheaper than the ~9.2% term loan, and it cuts the equity check by $400M. The trade-off is a fixed charge that rises every year for 20 years. EBITDA falls from an estimated $126M to $96M after rent, and leverage is 4.7x on debt alone but 5.5x once the lease is counted as debt at 8x rent.

### Debt Waterfall

Cash left after interest and 1% term loan amortization first repays the ABL, which is cleared by Year 3, and 75% of what remains then sweeps the term loan. Total debt falls from $450M to about $253M by Year 5, a 44% paydown, while EBITDA grows to about $145M, taking leverage from 4.7x to 1.7x.

### Returns and What the Fund Needs

Exiting in Year 5 at 7.25x EBITDA, roughly the entry multiple, gives an estimated 22.2% IRR and 2.7x MOIC, earned from EBITDA growth and debt paydown with no multiple expansion. The fund gets its money back at an exit of just 3.6x EBITDA and clears the typical 8% LP hurdle at 4.6x, so the downside cushion is wide. After a 20% carry, LPs would net about 19% before management fees.

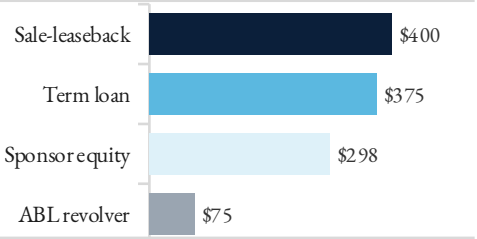
### Read-Across for the Sector

CRH keeps selling businesses outside its core building materials lines (it sold Oldcastle BuildingEnvelope to KPS for $3.45bn in 2022), and carve-outs like this are a steady source of deals for middle-market sponsors. Lawn and garden consumables are bought every season, which makes cash flow steadier than equipment sales, but GardenCore still carries the diesel and freight costs covered on page 2 and sells mainly through large home improvement retailers.

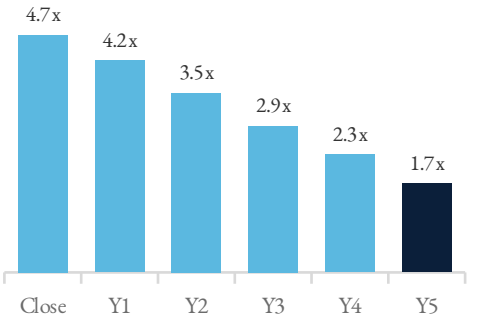
**Deal Terms**

| | |
|---|---|
| Announced | **30-Apr-2026** |
| Closed | **May 2026** |
| Enterprise value | **$1.1bn** |
| EV / EBITDAR (est.) | **8.7x** |
| Buyer | **Pacific Avenue (Fund II)** |
| Seller | **CRH plc** |
| Buy-side | **Jefferies, Weil, Sidley** |
| Sell-side | **BofA, Kilpatrick** |
| Sale-leaseback | **$400M, W. P. Carey** |
| Equity check (est.) | **$298M** |
| Base-case IRR (est.) | **22.2%** |

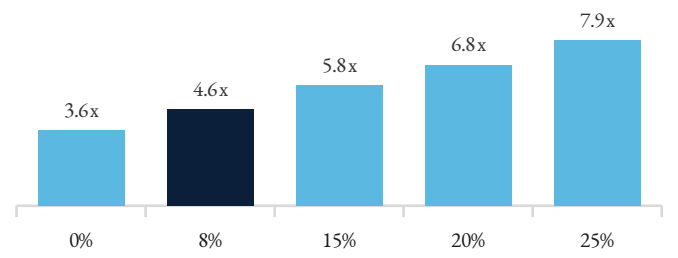
Sources of Funds ($M)




Total Debt / EBITDA (x)

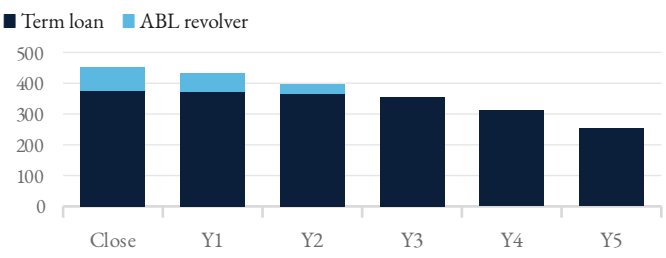



Exit Multiple Needed by Target IRR (x EBITDA)




*0% = money back; 8% = typical LP hurdle. Base case assumes a 7.25x exit in Year 5.*

Debt Outstanding by Facility ($M)




*Source: Pacific Avenue (Apr-2026); The Deal (May-2026); Sidley (May-2026); W. P. Carey 8-K and Q2-2026 earnings call; Business Wire (Aug-2025); CRH Q1-2026 results. Revenue, EBITDA, debt terms and returns are estimates from the accompanying GardenCore LBO model.*